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SUMMARY

Full-Stack & AI Engineer with 2+ years building production SaaS and AI systems across healthcare and recruitment tech. Specialized in multi-tenant architecture, RAG/LLM pipelines, and end-to-end delivery (Next.js, NestJS, PostgreSQL, Azure, GCP); experienced leading small teams and owning systems for hundreds of thousands of users.

EXPERIENCE

Full-Stack & AI Engineer (Agentic Full-Cycle Engineer)

MindHYVE.ai

Jan. 2026 — Present

- **Enterprise SaaS Architecture:** Architected and shipped ChironOM, a multi-tenant healthcare AI SaaS platform using Turborepo, pnpm, NestJS, Next.js 15, React 19, PostgreSQL, Drizzle ORM, Redis, and Azure Container Apps, covering clinical workflows for documentation, billing, RFA / UR authorization, and QME evaluation.
- **Production Ownership & Observability:** Maintained TheoAI, a live MindHYVE product serving 500k+ users; instrumented services with Azure Application Insights and LogRocket and act as the escalation point for production issues, tracing failures across services, logs, databases, and third-party APIs.
- **Data Layer & Tenant Isolation:** Re-architected multi-tenant isolation from a database-per-tenant model to shared-database PostgreSQL Row-Level Security with JWT-derived tenant context across 11 data domains, validated by automated isolation tests, and owned schema design, migrations, indexing, and transactional/row-level locking for race-free writes.
- **API & Auth Layer:** Built the platform's REST API in NestJS with JWT-derived tenant context, refresh-token rotation, and RBAC, plus Server-Sent Events for streaming AI responses to the client.
- **Multi-Agent AI Orchestration:** Designed a Multi-HYVE orchestration layer coordinating 5 specialized reasoning agents (medical, legal, financial, risk, ethics) across 6 clinical care pathways, invoked conditionally per case signals, with human-in-the-loop physician review.
- **Clinical RAG & AI Integration:** Delivered an MTUS / ACOEM RAG pipeline (PDF ingestion, section-aware chunking, HNSW vector search, two-step LLM retrieval), integrating Azure AI Foundry, DeepSeek-R1, Mistral Document AI OCR, Cohere embeddings, and pgvector.
- **Background Processing:** Built an event-driven asynchronous document pipeline on retrying BullMQ/Redis background workers (Azure Vision OCR with pdf-parse and Tesseract fallback) for OCR, structured extraction, and QME medical-record organization.
- **Testing, CI/CD & Compliance:** Wrote Jest unit, integration, and end-to-end tests, maintained Azure DevOps and GitHub Actions pipelines deploying to Azure Container Apps, and reinforced compliance with LLM evaluation harnesses, PHI audit logging, HIPAA risk assessment, STRIDE threat modeling, and Azure Bicep IaC.

Senior Full Stack Engineer (promoted from Full Stack Engineer)

QLU.ai

Sep. 2024 — Jan. 2026

- **Engineering Leadership:** Led the engineering for a business unit of three products (Dialer, Outreach automation, and AI Voice SDR), managing a team of 3 developers and 2 QA engineers across architecture, delivery, code quality, and release process.
- **Platform Architecture:** Directed architecture and delivery for 3 core platforms using Next.js, Node.js/Express, PostgreSQL, and production-grade backend services, scaling to 1,500+ users with 100+ concurrent sessions.
- **Cloud & Deployment:** Deployed and operated QLU's services on Google Cloud (Compute Engine, Cloud Run, Cloud Functions, Cloud Storage, Pub/Sub), using Cloud Logging for production debugging.
- **Billing & Data Migrations:** Refactored the credit and subscription systems end-to-end for 1,500+ users, and executed zero-downtime migrations of user, credit, and subscription data across PostgreSQL, MySQL, and MS SQL Server without interrupting production.
- **Outreach Automation & Caching:** Owned the end-to-end architecture for QLU's Outreach automation platform as an event-driven system with automation workflows, scheduling logic, and Redis-backed background jobs, and used Redis caching and pub/sub fan-out for real-time updates.
- **Dialer Platform:** Maintained and extended QLU's Twilio-based outbound Dialer (an AI-powered, high-volume calling platform for recruitment outreach), owning call reliability, call-state handling, status webhooks, real-time call events over WebSockets, and outbound-calling workflows.
- **AI People Search:** Delivered key features for QLU's core revenue product, AI People Search (natural-language candidate search over millions of profiles), supporting multi-million-dollar deal cycles, and optimized complex SQL and Elasticsearch queries to improve its performance by up to 90%.
- **Real-Time AI Voice SDR:** Engineered QLU's AI Voice SDR on top of the Dialer to autonomously place outbound recruitment calls, using NestJS, React, PostgreSQL, Twilio, and WebSockets for low-latency automated calling and conversation flows.

- **AI Agents MVP:** Architected and delivered the MVP for a new AI Agents application, rapidly building the frontend and backend architecture using React, Node.js, and Express.
- **APIs & Internal Tooling:** Built and maintained REST APIs in Node.js/Express and NestJS across search, billing, admin, and outreach domains, and shipped QLU’s admin operations panel for plan management, user blocking, and operational controls.
- **Authentication Microservice:** Developed a reusable authentication microservice using Node.js, Express, and PostgreSQL, migrating 1,500+ users from a monolithic database to improve system consistency.

Full Stack Engineer

Thy Reality
May 2024 — Sep. 2024

- **Crypto Property Trading Platform:** Built and shipped a cryptocurrency-based platform for buying and trading property shares, using NestJS, React, TypeScript, and Tailwind CSS, with secure purchase and trading workflows.
- **Mobile & GraphQL:** Developed the companion mobile app in React Native and designed optimized GraphQL APIs, improving system performance and real-time data handling.

React Developer Intern

Hexa IT
Jun. 2023 — Aug. 2023

- **Frontend Development:** Built reusable React components and user-facing features for production web applications, integrating RESTful APIs to backend services.

PROJECTS

NewsBiasDetect — Political Bias Detection in News (Final-Year Research)

FAST-NUCES
2023 — 2024

- **End-to-End NLP Research:** Designed and executed a research project on detecting political bias in Pakistani news (bias toward PTI, PML-N, PPP, or neutral), authored as a full IEEE-format research paper.
- **Dataset Construction:** Built and manually annotated a dataset of 1,000 political news articles with input from expert journalists, via news-API collection, scraping, cleaning, normalization, and preprocessing.
- **Transformer Fine-Tuning:** Fine-tuned and compared four transformer models: RoBERTa, BERT, DistilBERT, and DistilRoBERTa, using the Hugging Face Transformers library and PyTorch with mixed-precision (fp16) training on NVIDIA T4 GPUs.
- **Feature Engineering:** Combined textual features (media source, title, article body) with LDA topic modeling and lexical bias signals; the best configuration (RoBERTa) achieved 0.796 accuracy and 0.767 F1.
- **Evaluation:** Assessed all models on accuracy, precision, recall, and F1 over an 80/20 split, analyzing per-class confusion matrices to interpret model behavior.

Semantic Similarity in English Sentences

- Completed SemEval 2024 Task 1 on English Semantic Textual Relatedness (STR).
- Developed a system using a BERT-based model to accurately measure semantic similarity between English sentence pairs.

SKILLS

Languages	TypeScript, JavaScript, Python, SQL, HTML, CSS
Frameworks & Libraries	Next.js, React.js, React Native, Node.js, Express, NestJS, GraphQL, Tailwind CSS
Databases & Data	PostgreSQL, MySQL, MS SQL Server, MongoDB, Redis, Elasticsearch, pgvector, Drizzle ORM, Sequelize ORM
AI & ML	Python (PyTorch, Hugging Face Transformers, scikit-learn, pandas, NumPy, Gensim/LDA), RAG, LLM Integration, Vector Search, Embeddings, Fine-tuning, BERT / RoBERTa / Transformers, Multi-Agent Orchestration, NLP, Model Evaluation (Precision / Recall / F1)
Cloud & DevOps	DevOps Azure (Container Apps, AI Foundry, Bicep), GCP (Compute Engine, Cloud Run, Cloud Functions, Cloud Storage, Pub/Sub), Docker, Azure DevOps Pipelines, GitHub Actions, Turborepo, pnpm, BullMQ, Git
Testing & Observability	Jest (unit, integration, e2e), Azure Application Insights, LogRocket, Cloud Logging

EDUCATION

National University of Computer and Emerging Sciences Islamabad

2020 — 2024
CGPA 3.44

- Bachelor of Computer Science